similar to the buzzer circuit. When any of the switches (S1 through S8) is pressed, IC4 triggers for a preset time, and the appliance gets switched on for the preset time, and thereafter switches off automatically.

For proper understanding of the device, refer to Table 1, which indicates that the appliance will be enabled for a preset time when one switch (out of 8) is pressed momentarily.

An actual-size, single-side PCB layout for the optional appliance control circuit is shown in Fig. 5 and its component layout in Fig. 6. After assembling the circuit on PCB, enclose it in a suitable box. Interconnect CON2 and CON3 through male and female connectors. Now your appliance control device is ready to use. After powering on the circuit, press any switch among S1 through S8 from different room numbers (RN1 through RN8) to operate the appliance for a preset time.

Timer IC4 with RC components decides the time available at output pin 3 to switch on the appliance via relay RL1. Here in RC, R is equal to (R13 + VR1) and C is equal to C3 (1000µF). These time-deciding RC components can be set for 1 minute to 15 minutes, as shown in Table 2.

The 555 timer must be configured in monostable mode. The 555 timer IC4 starts timing when it triggers via switches S1 through S8 and switches on the appliance. Generally, the time duration for pin 3 of 555 timer IC will remain high. Its value can be derived from the relationship: T = 1.1xRxC

For making a 1-minute timer circuit, you can calculate the value of the resistor (VR1 + R13) to be around 55k. Similarly, for the 5-minute, 10-minute, and 15-minute timer circuits, the resistor values are 272.7k, 545.4k, and 818.2k, respectively. After the set preset time the appliance will switch off automatically.

After assembling the circuit, install the unit near the appliance to be controlled in such a way that when somebody from any room number (RN1 through RN8) presses their switch, the appliance gets switched on for the preset time.

Bonus. You can watch the video of the tutorial of this DIY project at *https://www.electronicsforu.com/videos-slideshows/live-diy-multi-switch-control-buzzer* EFY

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